

Siddharth Ray

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EDUCATION

University of California, Berkeley

May 2030

Bachelor of Science in Electrical Engineering and Computer Sciences

Vista del Lago High School

May 2026

High School Diploma, Valedictorian

GPA: 4.0 UW / 4.43 W

Patent: AI-based DJ system, pending

EXPERIENCE

FIRST Tech Challenge (FTC) Team #16236 Juice

Folsom, CA

Vice Captain, Software Division Lead

July 2021 – May 2026

- Developed advanced autonomous, localization, and robot-control systems, including a live relocalization system and sensor-fusion algorithms that achieved 0.1" average positional error.
- Led technical operations, recruitment, sponsorships, and outreach for a team ranked 9th of 8,000+ worldwide, reaching the 2023 FTC World Championship Division Finals and 2026 World Championship Playoffs; helped raise \$8K+/yr and increase rookie retention by 80%.

PROJECTS

Automated FTC Virtual Match Reconstruction and Analysis (Computer Vision) | [GitHub Repo](#)

April 2026

- Developed a lightweight FTC computer-vision pipeline that calibrated match broadcasts to field coordinates and tracked 4 robots through severe occlusion and collisions, enabling automated trajectory reconstruction from standard broadcast footage without GPU-intensive deep learning.
- Built end-to-end analytics tooling for competition scouting, including geometric tracking, merge/identity resolution, shot-event logging, and a local dashboard for calibration and batch processing, reducing manual scouting effort and enabling automated scouting on standard laptops without GPU acceleration.

Full-Stack Convention/Event Management System (Mobile/Desktop Applications)

April 2026

- Designed and developed a production event-management platform for a nonprofit, replacing legacy manual workflows with integrated ticketing, POS, meal management, scheduling, maps, and announcements.
- Built cross-platform applications with Flutter (Mobile), Electron.js (Desktop), and a Supabase/PostgreSQL backend, integrating Square payments, Firebase Cloud Messaging, Mapbox, role-based access, and real-time data synchronization.
- Implemented offline-first encrypted local storage for ticketing, check-in, and POS workflows, enabling reliable event operations during network outages and synchronizing transactions when connectivity was restored.

Ultra-Low Profile Backlit SmartFrame (Embedded Systems)

July 2026

- Designed and programmed low-power Arduino firmware with phototransistor-based automatic day/night activation and an RGB status-control interface.
- Engineered an isolated 12V/5V power-distribution system supporting up to 54W of high-density LED lighting within the enclosure's thermal and space constraints.
- Designed a 0.75-inch-thick enclosure optimizing component layout, heat dissipation, and light diffusion within a compact form factor with an accompanying discrete control panel

SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL, Dart, MATLAB

Frameworks & Software: OpenCV, PyTorch, Electron, Node.js, Mapbox, Jupyter, Git, PostgreSQL, Flutter, Supabase, Firebase

Hardware & Embedded: STM32, ESP32, Arduino, Raspberry Pi, Altium